

**HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
KHOA CÔNG NGHỆ THÔNG TIN**



DATABASE DESIGN TERM PROJECT

Database system for a construction company with multiple projects

Course: Database [E24-Class 5].

Team: 5.

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REQUIREMENT ANALYSIS

1. Company background and scenario

1.1 Overview

A34Build is a general contractor handling civil infrastructure, commercial buildings, and industrial facilities. The company runs multiple active projects at the same time across different sites. Each project has fixed deadlines, strict budgets, and safety requirements. Daily work involves field engineers, site managers, equipment operators, heavy machinery, specialized subcontractors, and tons of building materials.

The company currently relies on spreadsheets and paper logs. This causes scheduling delays, budget overruns, inventory loss at job sites, and slow cost tracking. The goal is to build a centralized relational database to manage projects, staff, equipment, materials, subcontracts, and site safety.

1.2 Main business operations

The database covers seven main operational areas:

1. Projects: Client contracts, job sites, phases, and task schedules.
2. Workforce: Job roles, reporting hierarchy, certifications, and task assignments.
3. Equipment: Owned and rented machinery, site deployments, run hours, and maintenance logs.
4. Materials: Suppliers, purchase orders, site deliveries, and stock levels.
5. Subcontractors: Trade specialties, contract agreements, and safety ratings.
6. Project costs: Budget tracking against spending on labor, equipment, materials, and subcontracts.
7. Safety: Safety audits, inspection scores, and incident reports.

2. Data requirements

2.1 Entity sets

Strong entity sets

1. DEPARTMENT: Company departments (Civil Engineering, Heavy Fleet, Procurement, Safety, Finance).
2. EMPLOYEE: Full-time and field staff (engineers, managers, operators, inspectors).
3. CLIENT: Public agencies and private developers that hire the company.
4. PROJECT: Construction contracts with defined budgets and delivery schedules.
5. PROJECT_SITE: Physical work locations for a project.
6. TASK: Work packages within a project phase assigned to workers.
7. EQUIPMENT: Heavy machinery and vehicles.
8. SUPPLIER: Vendors supplying raw materials and parts.
9. MATERIAL: Standard catalog items (cement, rebar, gravel, pipes).

10. PURCHASE_ORDER: Purchase orders sent to suppliers.
11. SUBCONTRACTOR: Outside contractors hired for specialized trades.
12. SAFETY_INSPECTION: Periodic site safety audits.
13. INCIDENT_REPORT: Logs of workplace accidents, equipment damage, or near-misses.

Weak entity sets

1. DEPENDENT: Family members of an employee for benefits and emergency contact.
 - Identifying entity: EMPLOYEE
 - Identifying relationship: HAS_DEPENDENT
 - Partial key: DependentName
2. PROJECT_PHASE: Stages within a project (Foundation, Structure, Finishing).
 - Identifying entity: PROJECT
 - Identifying relationship: CONSISTS_OF_PHASES
 - Partial key: PhaseNumber
3. PO_ITEM: Individual line items on a purchase order.
 - Identifying entity: PURCHASE_ORDER
 - Identifying relationship: CONTAINS_ITEM
 - Partial key: LineNumber
4. MAINTENANCE_RECORD: Service and repair logs for a machine.
 - Identifying entity: EQUIPMENT
 - Identifying relationship: HAS_MAINTENANCE
 - Partial key: RecordNumber

2.2 Attributes

Strong entity sets

1. DEPARTMENT

Attribute	Type	Description
DeptID	Primary Key	Unique department code (e.g., D01)
DeptName	Simple	Department name
OfficeLocation	Simple	Office room or floor
Budget	Simple	Annual operating budget

2. EMPLOYEE

Attribute	Type	Description
EmpID	Primary Key	Unique employee identifier
Name	Composite	Composed of FirstName, MiddleName, LastName

Address	Composite	Composed of Street, Ward, District, City
DOB	Simple	Date of birth
Age	Derived	Current date minus date of birth
HireDate	Simple	Employment start date
Salary	Simple	Monthly base salary
JobTitle	Simple	Job title (Site Engineer, Crane Operator)
Email	Candidate Key	Unique company email
PhoneNumber	Multi-valued	Contact phone numbers
Certifications	Multi-valued	Licenses and certificates (OSHA, Crane Class A)

3. CLIENT

Attribute	Type	Description
ClientID	Primary Key	Unique client identifier
ClientName	Simple	Organization or company name
ClientType	Simple	Sector (Government, Commercial, Residential, Industrial)
ContactPerson	Simple	Primary contact person
ContactEmail	Simple	Email address
ContactPhone	Multi-valued	Contact phone numbers
TaxCode	Candidate Key	Business tax code

4. PROJECT

Attribute	Type	Description
ProjectID	Primary Key	Project code (e.g., PRJ-2026-001)
ProjectName	Simple	Project name
StartDate	Simple	Planned start date
PlannedEndDate	Simple	Contractual deadline
ActualEndDate	Simple	Real completion date (null if ongoing)
ProjectDuration	Derived	Elapsed days from StartDate to ActualEndDate or today
EstimatedBudget	Simple	Contract budget limit
ActualCost	Derived	Total cost of labor, equipment, materials, and subcontracts
Status	Simple	Planned, In-Progress, Suspended, Completed

5. PROJECT_SITE

Attribute	Type	Description
SiteID	Primary Key	Unique site identifier
SiteName	Simple	Site name or sector

LocationCoordinates	Composite	Composed of Latitude, Longitude
Address	Composite	Composed of StreetAddress, District, City_Province
EnvironmentalPermit	Simple	Environmental permit number
SiteStatus	Simple	Operating state (Preparation, Active, Handed_Over)

6. TASK

Attribute	Type	Description
TaskID	Primary Key	Unique task identifier
TaskName	Simple	Task name (Pour pile B12)
EstimatedHours	Simple	Estimated work hours
ActualHours	Derived	Sum of worker hours logged on this task
ScheduledStartDate	Simple	Scheduled start date
ScheduledEndDate	Simple	Scheduled end date
ActualEndDate	Simple	Actual completion date (null if ongoing)
PriorityLevel	Simple	Low, Medium, High, Urgent
TaskStatus	Simple	Execution state (Pending, In_Progress, Completed, Blocked)

7. EQUIPMENT

Attribute	Type	Description
EquipmentID	Primary Key	Unique asset code
Model	Simple	Make and model (CAT 320D)
SerialNumber	Candidate Key	Manufacturer serial number
Category	Simple	Excavator, Crane, Truck, Paver, Mixer
Ownership	Simple	Owned or Rented
HourlyRate	Simple	Cost per operating hour
CurrentStatus	Simple	Available, Deployed, Under-Maintenance, Retired
TotalRunHours	Derived	Total operating hours across all jobs

8. SUPPLIER

Attribute	Type	Description
SupplierID	Primary Key	Unique vendor identifier
CompanyName	Simple	Registered company name
TaxCode	Candidate Key	Official business tax ID
ContactEmail	Simple	Sales email
ContactPhone	Multi-valued	Contact phone numbers

SupplyCategories	Multi-valued	Product lines (Steel, Cement, Lumber)
CreditRating	Simple	Rating grade (A, B, C)

9. MATERIAL

Attribute	Type	Description
MaterialID	Primary Key	Unique material SKU
MaterialName	Simple	Material name (Portland Cement PCB40)
Category	Simple	Material classification (Steel, Cement, Aggregate, Piping, Electrical)
StandardUnit	Simple	Unit of measurement (Ton, m3, Bag, Piece)
UnitPrice	Simple	Reference price per unit
MinStock	Simple	Minimum safety stock count

10. PURCHASE_ORDER

Attribute	Type	Description
PO_ID	Primary Key	Unique PO number
OrderDate	Simple	Date order was placed
ExpectedDeliveryDate	Simple	Promised arrival date
DeliveryStatus	Simple	Issued, Partial_Delivery, Fulfilled, Cancelled
TotalPOAmount	Derived	Sum of line subtotals

11. SUBCONTRACTOR

Attribute	Type	Description
SubcontractorID	Primary Key	Unique subcontractor identifier
CompanyName	Simple	Registered company name
TaxCode	Candidate Key	Official business tax ID
ContactEmail	Simple	Official contact email
ContactPhone	Simple	Primary dispatch phone number
TradeSpecialty	Simple	Specialized trade (Piling, Electrical, HVAC, Facade)
LicenseNumber	Candidate Key	Trade license number
SafetyRatingScore	Simple	Safety score (0.0 to 10.0)

12. SAFETY_INSPECTION

Attribute	Type	Description
InspectionID	Primary Key	Unique inspection log ID

InspectionDate	Simple	Date of inspection
InspectionType	Simple	Audit type (Routine, Unannounced, Pre-Operation)
InspectionScope	Simple	Inspected area or structure (e.g., Tower crane, Scaffold level 4)
OverallScore	Simple	Evaluation score (0 to 100)
InspectionResult	Simple	Audit conclusion (Passed, Conditional_Pass, Failed)
ViolationNoted	Simple	Safety issues found
FixDeadline	Simple	Deadline to fix issues (null if passed)

13. INCIDENT_REPORT

Attribute	Type	Description
IncidentID	Primary Key	Unique incident case ID
IncidentTimestamp	Simple	Date and time of incident
IncidentType	Simple	Event category (Fall, Electrical, Equipment_Collision, Falling_Object)
IncidentLevel	Simple	Impact classification (Warning, Minor, Serious, Fatal)
DamageDescription	Simple	Injury or equipment damage details
LostDays	Simple	Work days lost due to injury
Solution	Simple	Immediate remedial or preventive action taken

Weak entity sets

1. DEPENDENT

Attribute	Type	Description
DependentName	Partial Key	Full name of dependent
Gender	Simple	Gender (M, F, Other)
BirthDate	Simple	Date of birth
Relationship	Simple	Relationship to employee (Spouse, Child, Parent)
ContactPhone	Simple	Emergency contact phone number
IsEmergencyContact	Simple	Primary emergency contact flag (true or false)

2. PROJECT_PHASE

Attribute	Type	Description
PhaseNumber	Partial Key	Phase sequence number (1, 2, 3...)
PhaseTitle	Simple	Phase title (Excavation, Structure)
StartDate	Simple	Scheduled start date for this phase
TargetCompletionDate	Simple	Target completion date for phase

PhaseBudget	Simple	Budget allocated to this phase
PhaseStatus	Simple	Pending, Active, Under-Inspection, Accepted

3. PO_ITEM

Attribute	Type	Description
LineNumber	Partial Key	Line item number (1, 2, 3...)
QuantityOrdered	Simple	Quantity purchased
AgreedUnitPrice	Simple	Negotiated unit price
ItemSubtotal	Derived	QuantityOrdered multiplied by AgreedUnitPrice

4. MAINTENANCE_RECORD

Attribute	Type	Description
RecordNumber	Partial Key	Service record counter
ServiceDate	Simple	Date of service
MaintenanceType	Simple	Service type (Oil Change, Inspection, Hydraulic Repair, Overhaul)
Cost	Simple	Repair and parts cost
ServiceVendor	Simple	Workshop name or external vendor

3. Relationships and constraints

3.1 Relationships

1. DEPARTMENT_HEAD (EMPLOYEE to DEPARTMENT)

- Binary 1:1
- Participation: DEPARTMENT is total (every department has one manager); EMPLOYEE is partial (only some employees are managers).
- Rule: An employee manages at most one department.

2. WORKS_IN_DEPARTMENT (DEPARTMENT to EMPLOYEE)

- Binary 1:N
- Participation: EMPLOYEE is total (every employee must belong to one department); DEPARTMENT is total (every department has at least one employee).
- Rule: An employee belongs to exactly one department.

3. SUPERVISES (EMPLOYEE to EMPLOYEE)

- Unary (recursive) 1:N
- Participation: Both roles are partial (not all employees supervise, and the top director has no supervisor).
- Rule: An employee has at most one direct supervisor.

4. HAS_DEPENDENT (EMPLOYEE to DEPENDENT)

- Identifying 1:N
- Participation: EMPLOYEE is partial; DEPENDENT is total (a dependent must belong to an employee).

5. COMMISSIONS (CLIENT to PROJECT)

- Binary 1:N
- Participation: CLIENT is partial; PROJECT is total (each project must have a client).

6. DEPARTMENT_SPONSORS_PROJECT (DEPARTMENT to PROJECT)

- Binary 1:N
- Participation: DEPARTMENT is partial; PROJECT is total (each project is assigned to a managing department).

7. DIRECTS_PROJECT (EMPLOYEE to PROJECT)

- Binary 1:1
- Participation: PROJECT is total (every project has one site director); EMPLOYEE is partial.
- Rule: An employee directs at most one active project at a time.

8. HAS_SITES (PROJECT to PROJECT_SITE)

- Binary 1:N
- Participation: PROJECT is total (every project has at least one site); PROJECT_SITE is total.

9. CONSISTS_OF_PHASES (PROJECT to PROJECT_PHASE)

- Identifying 1:N
- Participation: PROJECT is total; PROJECT_PHASE is total.

10. CONTAINS_TASKS (PROJECT_PHASE to TASK)

- Binary 1:N
- Participation: PROJECT_PHASE is total; TASK is total.

11. PRECEDES (TASK to TASK)

- Unary (recursive) M:N
- Participation: Both roles are partial.
- Rule: Defines task order on the project schedule (for example, Foundation must finish before Framing starts).

12. WORKS_ON_TASK (EMPLOYEE to TASK)

- Binary M:N
- Attributes: AssignedDate, RoleOnTask, HoursLogged

- Participation: EMPLOYEE is partial; TASK is total (a task needs at least one assigned worker).

13. EQUIPMENT_DEPLOYMENT (EQUIPMENT to PROJECT_SITE)

- Binary M:N
- Attributes: DeploymentStartDate, DeploymentEndDate, HoursOperated, OperatorID (references EMPLOYEE)
- Participation: Both EQUIPMENT and PROJECT_SITE are partial.

14. HAS_MAINTENANCE (EQUIPMENT to MAINTENANCE_RECORD)

- Identifying 1:N
- Participation: EQUIPMENT is partial; MAINTENANCE_RECORD is total.

15. ISSUES_PO (PROJECT to PURCHASE_ORDER)

- Binary 1:N
- Participation: PROJECT is partial; PURCHASE_ORDER is total.

16. SUPPLIES_PO (SUPPLIER to PURCHASE_ORDER)

- Binary 1:N
- Participation: SUPPLIER is partial; PURCHASE_ORDER is total.

17. CONTAINS_ITEM (PURCHASE_ORDER to PO_ITEM)

- Identifying 1:N
- Participation: PURCHASE_ORDER is total; PO_ITEM is total.

18. ITEM_REFERENCES_MATERIAL (PO_ITEM to MATERIAL)

- Binary N:1
- Participation: PO_ITEM is total; MATERIAL is partial.

19. SITE_INVENTORY (PROJECT_SITE to MATERIAL)

- Binary M:N
- Attributes: CurrentStockQuantity, SafetyReorderQuantity, LastStocktakeDate
- Participation: Both are partial.

20. DELIVERY_DISPATCH (SUPPLIER x MATERIAL x PROJECT_SITE)

- Ternary relationship (M:N:P)
- Attributes: DeliveryCode, DeliveryDate, DeliveredQuantity, ReceiverEmployeeID (references EMPLOYEE), PO_ID (references PURCHASE_ORDER)
- Meaning: A supplier delivers a specific material to a project site against an approved purchase order on a given date.

21. ENGAGES_SUBCONTRACTOR (PROJECT to SUBCONTRACTOR)

- Binary M:N

- Attributes: ContractAgreementNo, ScopeDescription, ContractValue, WarrantyHoldRate, StartDate, CompletionDate
- Participation: Both are partial.

22. CONDUCTS_INSPECTION (PROJECT_SITE to SAFETY_INSPECTION to EMPLOYEE)

- PROJECT_SITE has 1:N with SAFETY_INSPECTION; EMPLOYEE has 1:N with SAFETY_INSPECTION.
- Participation: SAFETY_INSPECTION is total on both (each inspection records the site and the inspector).

23. LOGS_INCIDENT (PROJECT_SITE to INCIDENT_REPORT)

- Binary 1:N
- Participation: PROJECT_SITE is partial; INCIDENT_REPORT is total (each incident happens at a site).

24. INVOLVES_EMPLOYEE (EMPLOYEE to INCIDENT_REPORT)

- Binary 1:N
- Participation: EMPLOYEE is partial; INCIDENT_REPORT is partial (an incident may involve an injured worker, or may solely involve property/equipment damage).
- Rule: An incident report references at most one primary affected employee.

4. Business rules

4.1 Domain constraints

- Amounts must be positive (> 0): Salary, Budget, PhaseBudget, ContractValue, UnitPrice, AgreedUnitPrice, Cost, HourlyRate.
- Quantities and hours must be non-negative (≥ 0): QuantityOrdered, DeliveredQuantity, CurrentStockQuantity, MinStock, EstimatedHours, ActualHours, HoursLogged, HoursOperated, LostDays.
- Status and classification fields use fixed values:
 - PROJECT.Status: Planned, In-Progress, Suspended, Completed
 - PROJECT_SITE.SiteStatus: Preparation, Active, Handed_Over
 - PROJECT_PHASE.PhaseStatus: Pending, Active, Under-Inspection, Accepted
 - TASK.PriorityLevel: Low, Medium, High, Urgent
 - TASK.TaskStatus: Pending, In_Progress, Completed, Blocked
 - EQUIPMENT.Ownership: Owned, Rented
 - EQUIPMENT.CurrentStatus: Available, Deployed, Under-Maintenance, Retired
 - PURCHASE_ORDER.DeliveryStatus: Issued, Partial_Delivery, Fulfilled, Cancelled
 - SAFETY_INSPECTION.InspectionResult: Passed, Conditional_Pass, Failed
 - INCIDENT_REPORT.IncidentLevel: Warning, Minor, Serious, Fatal

- INCIDENT_REPORT.IncidentType: Fall, Electrical, Equipment_Collision, Falling_Object

4.2 Date and time rules

- End dates cannot be earlier than start dates ($\text{StartDate} \leq \text{PlannedEndDate}$, $\text{StartDate} \leq \text{ActualEndDate}$).
- Workers must be at least 18 years old ($\text{CurrentDate} - \text{DOB} \geq 18$ and $\text{HireDate} \geq \text{DOB} + 18$).
- Task start and end dates must fall within the dates of the parent project phase.

4.3 Rules not modeled in ERD

These rules cannot be shown with ER notation and will be enforced through triggers, check constraints, or application logic:

1. Phase budget limit: The sum of all phase budgets in a project cannot exceed the project's estimated budget.
2. No overlapping machine bookings: A piece of equipment cannot be scheduled at two different sites during overlapping dates.
3. Operator licensing: An employee assigned to run heavy equipment must have the required license listed in their certifications.
4. Daily work limit: A worker cannot log more than 12 hours total across all tasks in a single day.
5. No circular task dependencies: The PRECEDES relationship between tasks must not create cycles.
6. Stock limit on consumption: Tasks cannot consume more material than the current stock recorded at that site.
7. Single project limit for directors: An employee can be the director of only one active project at a time.
8. Accident emergency notification: When an incident involves an employee, the system must trigger a notification looking up their emergency contact in DEPENDENT where $\text{IsEmergencyContact} = \text{true}$.

5. Database applications

5.1 Data loading

- Import the standard material catalog with units and benchmark prices.
- Add new employees, job titles, and verified licenses.
- Set up new project contracts, site locations, and milestone schedules.
- Register newly purchased or rented machinery with serial numbers.

5.2 Updates

- Log daily worker hours on specific tasks.

- Record material deliveries arriving at a site and update on-site inventory counts.
- Deduct materials used during daily work from site inventory.
- Update equipment operating hours and switch status to maintenance when service intervals arrive.
- Mark project phases as completed after inspection sign-off.
- Log site safety audit scores and incident reports.

5.3 Queries

- Find heavy equipment by category that is available and not booked for the next two weeks.
- Check which materials at a specific site have dropped below their minimum safety stock (MinStock).
- List urgent tasks that are past their scheduled end date.
- Look up the reporting chain of an employee up to the general director.
- View active subcontractors on a project with their contracted amounts and remaining warranty hold balances (WarrantyHoldRate).

5.4 Reports

- Cost vs. budget variance: Compares the initial project budget against total spending on labor, equipment, materials, and subcontracts.
- Earned value analysis: Computes Planned Value (PV), Earned Value (EV), Actual Cost (AC), Cost Performance Index (CPI), and Schedule Performance Index (SPI).
- Site safety report: Summarizes audit scores, open safety issues, lost work days (LostDays), and total injury-free hours per site.
- Subcontractor scorecard: Ranks subcontractors by delivery speed, work quality, and safety compliance.
- Fleet utilization report: Shows machine run hours (TotalRunHours) against idle hours and hourly rate costs (HourlyRate).

Entity-Relationship Diagram

This section presents the ERD of the A34Build database. Figure 1 shows the whole model on a single page, with every entity set, relationship, and attribute. The six domains (Organization, Projects, Equipment, Materials and procurement, Subcontractors, and Safety) are described below the figure. The diagram covers the entity sets in Section 2.1, the attributes in Section 2.2, and the relationships in Section 3.1.

Notation: rectangles are strong entity sets and double rectangles are weak entity sets. Diamonds are relationships, and double diamonds are identifying relationships. Ellipses are attributes: underlined for keys, double-lined for multi-valued attributes, and dashed for derived attributes. A double line means total participation. Cardinalities are marked 1, N, M, and P. An aggregation is a relationship that takes part in another relationship, so it is drawn as an ordinary diamond linked to that relationship.

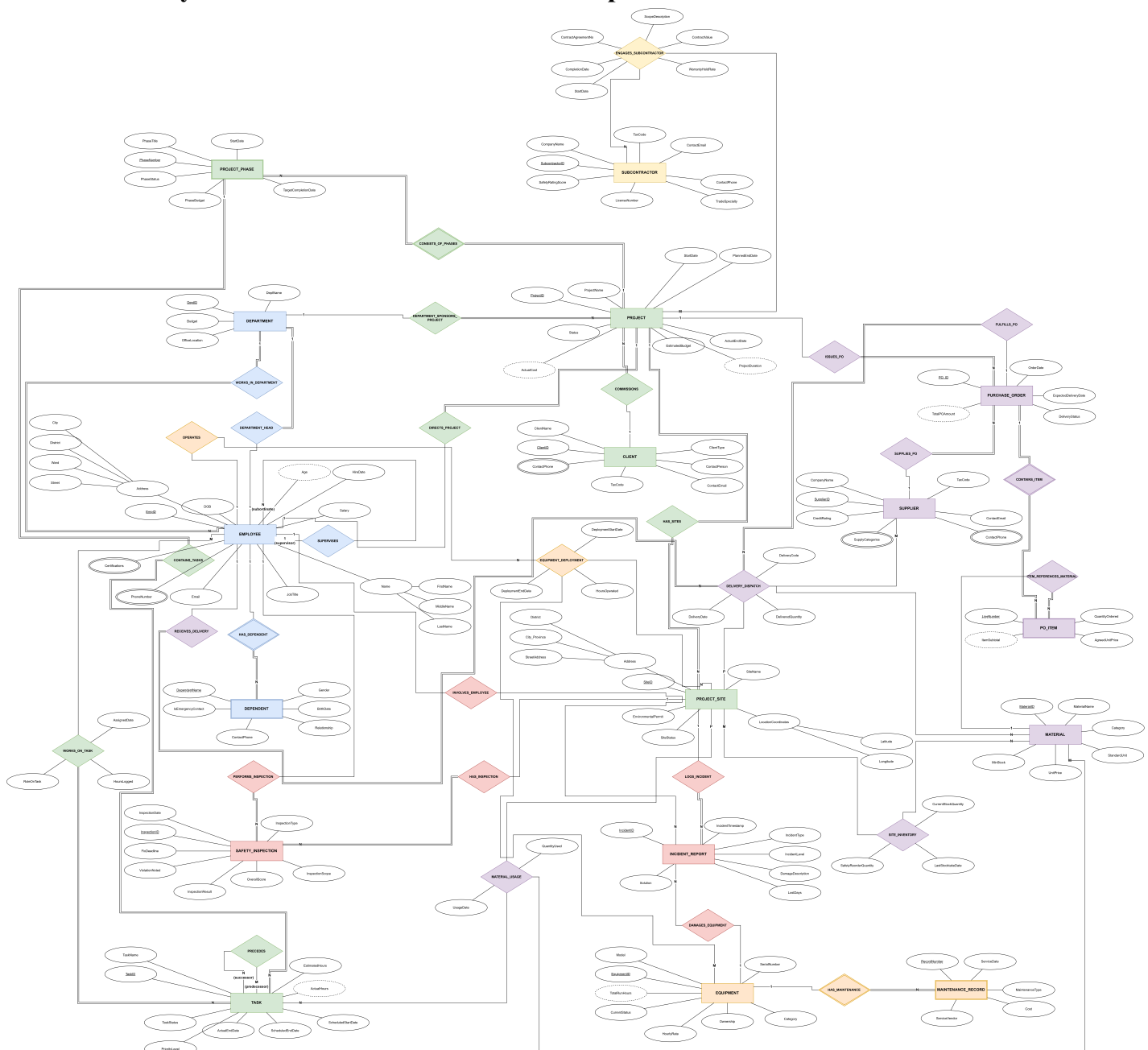


Figure 1. ERD of the A34Build database. Complete model on a single page, with all entity sets, relationships, and attributes. Entity sets and relationships are colour-coded by domain: blue for Organization, green for Projects, purple for Materials, orange for Equipment, yellow for

Subcontractors, and red for Safety. DELIVERY_DISPATCH and EQUIPMENT_DEPLOYMENT are aggregations: relationships treated as entities so they can take part in further relationships (FULFILLS_PO, RECEIVES_DELIVERY, and OPERATES).

Organization. DEPARTMENT, EMPLOYEE, and DEPENDENT. DEPARTMENT_HEAD (1:1) and WORKS_IN_DEPARTMENT (1:N) link employees to departments, SUPERVISES is the recursive reporting line, and HAS_DEPENDENT is the identifying relationship for the weak entity set DEPENDENT. Name and Address are composite, Age is derived, and PhoneNumber and Certifications are multi-valued.

Projects. CLIENT, PROJECT, PROJECT_SITE, PROJECT_PHASE, and TASK. A client commissions many projects, and each project has sites, phases, a managing department, and one site director. Phases contain tasks, PRECEDES orders tasks within the schedule, and WORKS_ON_TASK (M:N) records AssignedDate, RoleOnTask, and HoursLogged for each employee. ProjectDuration, ActualCost, and ActualHours are derived attributes.

Equipment. EQUIPMENT and its weak entity set MAINTENANCE_RECORD, linked by HAS_MAINTENANCE. EQUIPMENT_DEPLOYMENT is an M:N relationship between EQUIPMENT and PROJECT_SITE with DeploymentStartDate, DeploymentEndDate, and HoursOperated. It is an aggregation so that OPERATES can tie each deployment to the employee who runs the machine. TotalRunHours is derived, and SerialNumber is a candidate key.

Materials and procurement. SUPPLIER, PURCHASE_ORDER, PO_ITEM, and MATERIAL. A project issues purchase orders, a supplier fulfils them, and each order contains line items that reference a material. SITE_INVENTORY holds stock levels per site and material. DELIVERY_DISPATCH is the ternary relationship among SUPPLIER, MATERIAL, and PROJECT_SITE; FULFILLS_PO ties each delivery to its purchase order, and RECEIVES_DELIVERY records the employee who accepts it. MATERIAL_USAGE records the quantity of a material used by a task at a site.

Subcontractors. SUBCONTRACTOR and PROJECT are linked by the M:N relationship ENGAGES_SUBCONTRACTOR, which carries ContractAgreementNo, ScopeDescription, ContractValue, WarrantyHoldRate, StartDate, and CompletionDate (Section 3.1). A subcontractor can work on many projects, and a project can engage many subcontractors. LicenseNumber is a candidate key of SUBCONTRACTOR.

Safety. SAFETY_INSPECTION and INCIDENT_REPORT. Each inspection belongs to one project site (HAS_INSPECTION) and one inspecting employee (PERFORMS_INSPECTION). Each incident is logged at a site (LOGS_INCIDENT) and can involve an employee (INVOLVES_EMPLOYEE) or damage a piece of equipment (DAMAGES_EQUIPMENT).

Relational Schemas

This section presents the relational schemas derived from the ERD. Figure 2 shows the whole schema on a single page, with every column of all 29 tables. The six domains (Organization, Projects, Equipment, Materials and procurement, Subcontractors, and Safety) are described below the figure. The schemas follow the entity sets in Section 2.1, the attributes in Section 2.2, and the relationships in Section 3.1.

Notation: each box is a table, with the table name in the header and one cell per column. Underlined columns form the primary key, and a dashed underline marks the partial key of a weak entity set. Arrows run from a foreign key to the primary key it references and take the colour of the referencing table; a composite foreign key carries a label listing its columns. Colours match the ERD: blue for Organization, green for Projects, purple for Materials, orange for Equipment, yellow for Subcontractors, and red for Safety. Composite attributes are split into their components, multi-valued attributes become separate tables, and derived attributes are not stored.

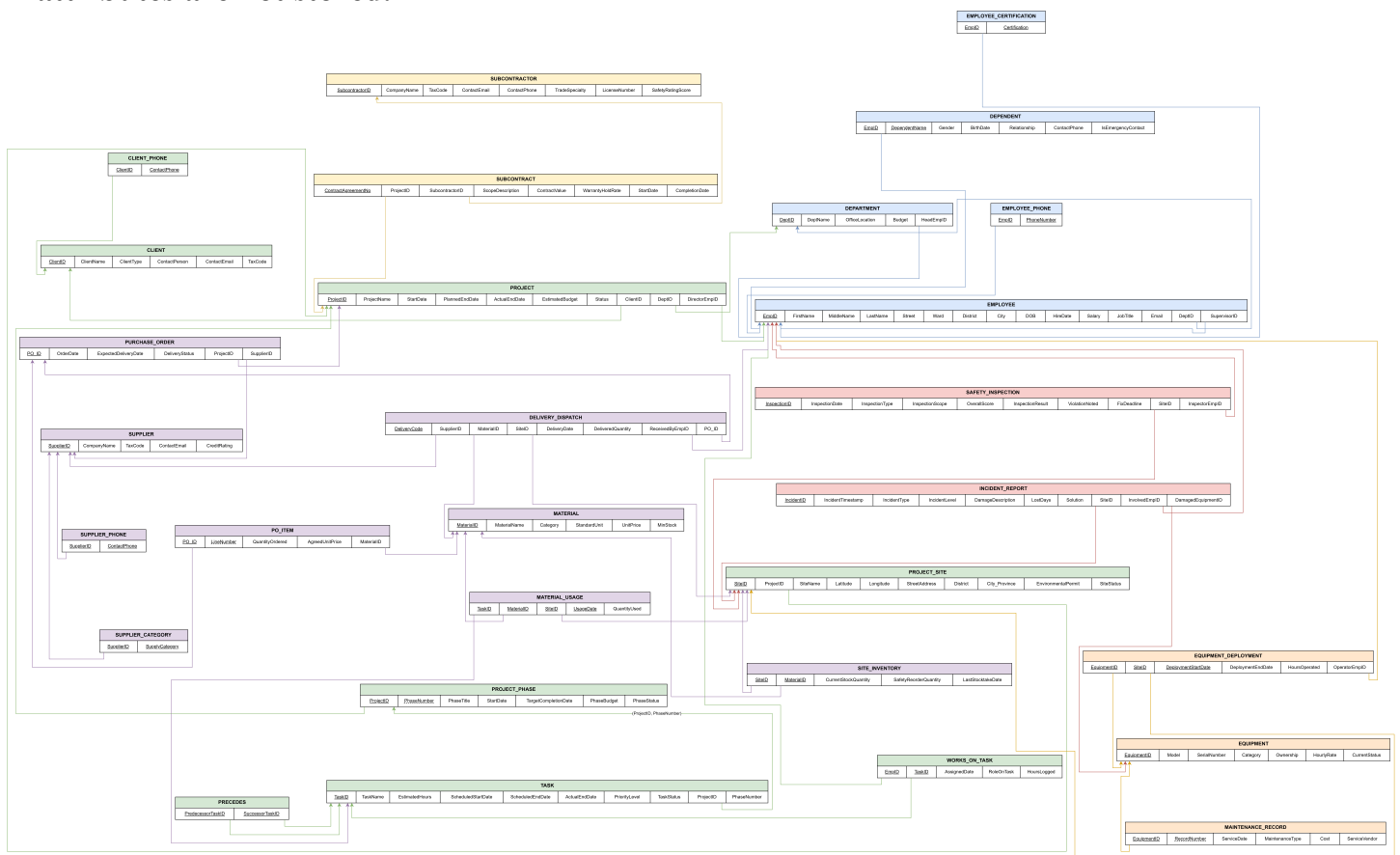


Figure 2. Relational schema. All 29 tables on a single page, with every column. Underlined columns form the primary key, a dashed underline marks a partial key, and each arrow points from a foreign key to the primary key it references. The composite foreign key of TASK is labelled (ProjectID, PhaseNumber).

Organization. DEPARTMENT, EMPLOYEE, EMPLOYEE_PHONE, EMPLOYEE_CERTIFICATION, and DEPENDENT. Name and Address are split into FirstName, MiddleName, LastName and Street, Ward, District, City. The multi-valued attributes PhoneNumber and Certifications become their own tables, keyed by EmpID plus the value. DEPENDENT takes its key (EmpID, DependentName) from its owner EMPLOYEE. HeadEmpID in DEPARTMENT, and DeptID and SupervisorID in EMPLOYEE, are foreign keys.

Projects. CLIENT, CLIENT_PHONE, PROJECT, PROJECT_SITE, PROJECT_PHASE, TASK, PRECEDES, and WORKS_ON_TASK. PROJECT references CLIENT, DEPARTMENT, and EMPLOYEE (DirectorEmpID). PROJECT_PHASE is keyed by (ProjectID, PhaseNumber), and TASK references it through the composite foreign key (ProjectID, PhaseNumber). PRECEDES and WORKS_ON_TASK hold the recursive and M:N relationships among tasks and employees.

Equipment. EQUIPMENT, MAINTENANCE_RECORD, and EQUIPMENT_DEPLOYMENT. MAINTENANCE_RECORD is keyed by (EquipmentID, RecordNumber). EQUIPMENT_DEPLOYMENT is keyed by (EquipmentID, SiteID, DeploymentStartDate), and OperatorEmpID references EMPLOYEE.

Materials and procurement. SUPPLIER, SUPPLIER_PHONE, SUPPLIER_CATEGORY, MATERIAL, PURCHASE_ORDER, PO_ITEM, SITE_INVENTORY, DELIVERY_DISPATCH, and MATERIAL_USAGE. The multi-valued ContactPhone and SupplyCategories of SUPPLIER become separate tables. PO_ITEM is keyed by (PO_ID, LineNumber), and SITE_INVENTORY by (SiteID, MaterialID). DELIVERY_DISPATCH has its own key DeliveryCode and references SUPPLIER, MATERIAL, PROJECT_SITE, EMPLOYEE (ReceivedByEmpID), and PURCHASE_ORDER (PO_ID). MATERIAL_USAGE is keyed by (TaskID, MaterialID, SiteID, UsageDate).

Subcontractors. SUBCONTRACTOR and SUBCONTRACT. The M:N relationship ENGAGES_SUBCONTRACTOR between PROJECT and SUBCONTRACTOR becomes the table SUBCONTRACT, keyed by ContractAgreementNo, with ProjectID and SubcontractorID as foreign keys. ScopeDescription, ContractValue, WarrantyHoldRate, StartDate, and CompletionDate are the relationship attributes listed in Section 3.1.

Safety. SAFETY_INSPECTION and INCIDENT_REPORT. Both tables carry SiteID as a foreign key to PROJECT_SITE. SAFETY_INSPECTION also holds InspectorEmpID (EMPLOYEE), while INCIDENT_REPORT holds InvolvedEmpID (EMPLOYEE) and DamagedEquipmentID (EQUIPMENT).